

RESEARCH INTERESTS

I study **efficient, decision-aware LLM systems** by turning gaps between human judgment and agent behavior into runtime control. My work asks when an LLM system should retrieve, read, verify, stop, or route work to a smaller model. I study these decisions so reliable behavior remains practical under real deployment constraints: (i) **adaptive inference & runtime control** — deciding when an agent should retrieve, read, verify, or stop (HALT, READ-GATE); (ii) **evaluation beyond a single score** — trajectory-level failure decomposition, measuring accuracy and cost together; (iii) **distillation & amortization** — moving repeated LLM work offline into lightweight surrogates so the serving path stays LLM-free, controlled, and verifiable.

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology)

M.S. in Data Science · GPA 3.65/4.3 (93.5/100) · Advisor: Prof. Mun Yong Yi

Daejeon, Korea

Sep. 2022 – Aug. 2024

Handong Global University

B.A. in Economics & B.S. in Information and Communication Technology

GPA 3.59/4.5 (Major 3.88/4.5)

Pohang, Korea

Mar. 2018 – Aug. 2022

PUBLICATIONS

Selected Publications

* denotes equal contribution

One additional co-first-author manuscript (recommendation) is in preparation for resubmission; title available upon request.

[1] **HALT: Verification-Aware Stopping for Retrieval-Augmented Search Agents.**

Daeyoung Roh, Donghee Han. *Findings of EMNLP 2026*. arXiv:2608.02009.

Stops a frozen search agent on *evidence coverage* rather than generator confidence — up to 45% fewer search loops with clean hop targets; the deployable setting passes EM non-inferiority on all three benchmarks.

[2] **Before Reasoning Can Fail: Pre-Evidence Procedural Failures in Agentic RAG.**

Daeyoung Roh, Donghee Han. *Preprint*. arXiv:2608.02011.

Trace-level decomposition of 12,000 trajectories; a runtime read-gate recovers up to +9.4 points without changing the model.

[3] **CoVeR: Coverage-Based Routing of Verifier Calls in Agentic Retrieval.**

Daeyoung Roh, Donghee Han. *Preprint*. PDF.

Routes per-step verifier calls with a frozen-embedding coverage margin, cutting 62–68% of calls at TOST-equivalent accuracy.

Additional Peer-Reviewed Publications

[4] **LiteUP4Rec: Efficient User Profiling from LLM-Derived Semantic Supervision.**

Donghee Han*, Daeyoung Roh*, Jiwon Jeong, Mun Yong Yi. *CIKM 2026*.

Amortizes LLM-based user profiling into a sequence-conditioned surrogate: profile refresh 641 / 42.9 ms → 0.113 ms, GPU 12.5 → 0.48 GB, +40% avg. accuracy.

[5] **Distilling LLM Reasoning into Dense Encoders: Bridging the Accuracy-Efficiency Gap in Recommendation.**

Donghee Han, Daeyoung Roh, A Young Kim, Hwanjun Song, Mun Yong Yi. *Findings of ACL 2026*.

Distills LLM reasoning into a 0.3B dense encoder via oracle-guided supervision; the LLM-free student outperforms its 12B teacher while cutting inference latency by an order of magnitude.

[6] **Fine-Grained Multi-Prompt Essay Scoring with Multi-Level Disentanglement.**

Donghee Han, Daeyoung Roh, Euilwan Han, Hwanjun Song, Mun Yong Yi.

Data Mining and Knowledge Discovery (ECML-PKDD Journal Track), 2025.

[7] **CTIP: Towards Accurate Tabular-to-Image Generation for Tire Footprint Generation.**

Daeyoung Roh, Donghee Han, Jihyun Nam, Youngbin You, Jeongheon Park, Jungsoo Oh, Mun Yong Yi.

IEEE/CVF WACV 2025.

[8] **Closer through Commonality: Enhancing Hypergraph Contrastive Learning with Shared Groups.**

Daeyoung Roh, Donghee Han, Daehee Kim, Keejun Han, Mun Yong Yi.

IEEE International Conference on Big Data 2025.

[9] **SIG-Net: GNN-based Dropout Prediction in MOOCs Using Student Interaction Graph.**

Daeyoung Roh, Donghee Han, Daehee Kim, Keejun Han, Mun Yong Yi. *ACM/SIGAPP SAC 2024*.

INDUSTRY EXPERIENCE

Ensapia (formerly Cocone), AI Lab

AI Engineer

Seoul, Korea

Jul. 2024 – Present

- Designed and operate a production **multi-agent RAG system** for enterprise knowledge search over heterogeneous sources (documents, internal portals, Google Workspace), serving diverse questions across departments; built a **feedback-driven knowledge-update pipeline**. Observations of hallucination and multi-hop failures in this system directly motivated my first-author research on agent runtime control.
- Built an **HR AI analysis platform** (STT+LLM interview evaluation) deployed in production for Korean and Japanese HR teams, and a **legal AI compliance tool** combining image-similarity search with LLM reasoning for the legal team.
- Led the **design-generation platform**: fine-tuning, inference serving, and operations for Flux-based image generation on AWS EKS / RunPod.

RESEARCH EXPERIENCE

KAIST Knowledge Innovation Research Center (KIRC)

Project Manager, Industry Collaboration with Hankook Tire

Daejeon, Korea

Nov. 2023 – Jun. 2024

- Led tabular-to-image generation for tire footprint prediction; developed CTIP (tabular representation learning + Stable Diffusion), published at WACV 2025.

KAIST Knowledge Innovation Research Center (KIRC)

Project Manager, Veterinary Diagnosis Prediction

Daejeon, Korea

Jul. 2023 – Dec. 2023

- Built tree-ensemble models predicting ~20 canine diagnoses from blood tests under small, imbalanced data.

Fount Inc., AI Core Department

Research Intern — ML for stock-price prediction

Seoul, Korea

Jul. 2021 – Aug. 2021

Handong Global University

Undergraduate Research Assistant

Pohang, Korea

2021 – 2022

- Systemic-risk monitoring for pooled investment products — return-series preprocessing and *conditional value-at-risk* spillover estimation by product category (Financial Supervisory Service project); ML-based regional housing-market analysis and fraud detection (Korea Real Estate Board project).

HONORS & AWARDS

- **Excellence Award**, Hankook & Company Group–KAIST AI Competition, 2023 — tire-footprint generation; developed into the WACV 2025 paper.
- **Creativity Award**, K-DS Conference Hackathon, 2023 — RAG-based Korean-history chatbot with a thesaurus-grounded vector DB.

TEACHING & LEADERSHIP

- Teaching Assistant for four courses, Handong Global University (2021): Macroeconomics, Econometrics, Intro. to Economics, Intro. to Data Application.
- Editor-in-Chief (2019; Staff Reporter, 2018), *Handong Press* — published the student newspaper *Handong Paper*; led ~20 student reporters: writing training, budget management, print/video editorial planning.

TECHNICAL SKILLS

- **ML / LLM systems**: PyTorch, HuggingFace, vLLM, LangChain, LoRA fine-tuning, retrieval (BM25/dense/FAISS), agentic RAG.
- **Statistical analysis**: paired bootstrap, non-inferiority & TOST equivalence testing, Holm correction, McNemar / permutation tests.
- **Infrastructure**: AWS EKS, RunPod, Docker, PostgreSQL, FastAPI, GPU serving pipelines.